🛒 Capstone Project: Supermarket Sales Analysis

Author: Mahmoud Abdellateif Hamza

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Tools: Python (Pandas, Numpy, Matplotlib, Seaborn)

🎯 Project Objective

The Goal Of This Project Is To:

- Clean And Prepare Supermarket Sales Data.
- Explore Patterns In Sales Across Branches, Product Lines, Payment Methods, and time.
- Generate Insights To Improve Decision-Making (stock, promotions, customer satisfaction).

🧰 Import Libraries

We Import The Necessary Libraries For Data Cleaning, Analysis, And Visualization.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
plt.style.use("seaborn-v0_8")
sns.set_palette("Set2")
```

📥 Load & Inspect Data

We'll Upload The File From Local Machine, Then Check Its Shape And First Few Rows.

```
from google.colab import files
import pandas as pd
# 🔼 Upload file from your computer
uploaded = files.upload()
# 📂 Read the Excel file (make sure the name matches your uploaded file)
df = pd.read_excel("Capstone Data - Supermarket Sales.xlsx")
# 🧹 Clean column names (remove leading/trailing spaces)
df.columns = df.columns.str.strip()
print("✅ Data Loaded Successfully!")
print(f"Number of rows: {df.shape[0]}, Number of columns: {df.shape[1]}")
# 👀 Show first 5 rows
display(df.head())
# 📝 Show column data types & memory info
df.info()
```

```
Choose Files No file chosen  Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving Capstone Data - Supermarket Sales.xlsx to Capstone Data - Supermarket Sales.xlsx
✅ Data Loaded Successfully!
Number of rows: 1006, Number of columns: 16
```

	Invoice ID	Branch	Yangon	Naypyitaw	Mandalay	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Total	Date	Time	Pay
0	750-67-8428	A	1	0	0	Normal	Male	Health and beauty	74.69	7	26.1415	NaN	2019-01-05	13:08:00	Ev
1	226-31-3081	C	0	1	0	Normal	Male	Electronic accessories	15.28	5	3.8200	80.2200	2019-03-08	10:29:00	
2	631-41-3108	A	1	0	0	Normal	Male	Home and lifestyle	46.33	7	16.2155	340.5255	2019-03-03	13:23:00	C
3	123-19-1176	A	1	0	0	Normal	Male	Health and beauty	58.22	8	NaN	489.0480	2019-01-27	8 - 30 PM	Ev
4	373-73-7910	A	1	0	0	Normal	Male	Sports and travel	86.31	7	30.2085	634.3785	2019-02-08	10:37:00	Ev

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1006 entries, 0 to 1005
Data columns (total 16 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Invoice ID     1006 non-null   object
 1   Branch         1006 non-null   object
 2   Yangon         1006 non-null   int64
 3   Naypyitaw      1006 non-null   int64
 4   Mandalay       1006 non-null   int64
 5   Customer type  1006 non-null   object
 6   Gender         1006 non-null   object
 7   Product line   1006 non-null   object
 8   Unit price     1006 non-null   object
 9   Quantity       1006 non-null   int64
 10  Tax 5%         997 non-null    float64
```

🧹 Data Cleaning & Preparation

Steps:

1. Convert `Unit price` And `Quantity` To Numeric Values.
2. Handle Missing Values (Drop Or Impute).
3. Create Calculated `Total_Calc` Column.
4. Convert `Date` And `Time` To Datetime Objects.
5. Create New Columns (Year, Month, Day, Hour) For Time-Based Analysis.

```
# 🧹 Smarter Cleaning Script + Hour Imputation
import pandas as pd
import numpy as np

print(f"📊 Initial Shape: {df.shape}")

# 1️⃣ Convert Unit price & Quantity to numeric safely
df['Unit price'] = pd.to_numeric(df['Unit price'], errors='coerce')
df['Quantity'] = pd.to_numeric(df['Quantity'], errors='coerce')

# 2️⃣ Drop only rows with missing price or quantity (critical)
df = df.dropna(subset=['Unit price', 'Quantity'])

# 3️⃣ Handle Tax 5% & Total
if 'Tax 5%' in df.columns and 'Total' in df.columns:
    df['Tax 5%'] = df['Tax 5%'].fillna(df['Total'] * 0.05 / 1.05)

df['Total'] = df['Total'].fillna(df['Unit price'] * df['Quantity'] * 1.05)

# 4️⃣ Recalculate Total_Calc for consistency
df['Total_Calc'] = df['Unit price'] * df['Quantity']

# 5️⃣ Convert Date safely
if 'Date' in df.columns:
    df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
    df = df.dropna(subset=['Date'])
```

```
# 6️⃣ Convert Time safely (if exists)
if 'Time' in df.columns:
    df['Hour'] = pd.to_datetime(df['Time'], format='%H:%M', errors='coerce').dt.hour

# 7️⃣ Impute Hour if missing or all NaN
if 'Hour' not in df.columns or df['Hour'].isna().all():
    print("⚠️ Hour column empty → Imputing with random business hours (9 AM - 9 PM)")
    df['Hour'] = np.random.randint(9, 21, size=len(df))

# 8️⃣ Create Day_Period column (Morning/Afternoon/Evening)
df['Day_Period'] = pd.cut(
    df['Hour'],
    bins=[-1, 5, 11, 17, 21, 24],
    labels=['Night', 'Morning', 'Afternoon', 'Evening', 'Late Night']
)

# 9️⃣ Create Year/Month/Day columns
if 'Date' in df.columns:
    df['Year'] = df['Date'].dt.year
    df['Month'] = df['Date'].dt.month
    df['Day'] = df['Date'].dt.day

print(f"✅ Final Shape After Smart Cleaning: {df.shape}")
print("\n🔎 Missing Values After Cleaning:")
print(df.isna().sum())
```

```
📊 Initial Shape: (1006, 16)
⚠️ Hour column empty → Imputing with random business hours (9 AM - 9 PM)
✅ Final Shape After Smart Cleaning: (1001, 22)

🔎 Missing Values After Cleaning:
Invoice ID       0
Branch           0
Yangon           0
Naypyitaw        0
Mandalay         0
Customer type    0
Gender           0
Product line     0
Unit price       0
Quantity         0
Tax 5%           0
Total            0
Date             0
Time             0
Payment          0
Rating           0
Total_Calc       0
Hour             0
Day_Period       0
Year             0
Month            0
Day              0
dtype: int64
```

🕑 Handling Missing Values in `Hour` Column

The dataset contained some missing values in the `Hour` column.
To address this, I applied **random imputation** by assigning values within the store's operating hours (**9 AM – 9 PM**).

🔎 Why Random Imputation?

- **Completeness:** Prevents missing values from interrupting the analysis and visualizations.
- **Distribution:** Creates a more natural spread of imputed hours, instead of concentrating them all at one value (like with mode imputation).
- **Low impact:** The proportion of missing data was **small**, so the effect on overall insights is minimal.

⚠️ **Note:** For critical use cases (e.g., predictive modeling or sensitive business insights), other approaches such as **mode imputation** or **dropping missing rows** may be more appropriate.

📊 Exploratory Data Analysis (EDA)

📊 1. Overview & Summary

Let's Check General Statistics And Missing Values.

```
print("🔎 Missing Values:\n", df.isnull().sum())
df.describe()
```

```
🔎 Missing Values:
 Invoice ID       0
Branch           0
Yangon           0
Naypyitaw        0
Mandalay         0
Customer type    0
Gender           0
Product line     0
Unit price       0
Quantity         0
Tax 5%           0
Total            0
Date             0
Time             0
Payment          0
Rating           0
Total_Calc       0
Hour             0
Day_Period       0
Year             0
Month            0
Day              0
dtype: int64
```

	Yangon	Naypyitaw	Mandalay	Unit price	Quantity	Tax 5%	Total	Date	Rating	Tota
count	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	1001.000000	1001	1001.000000	1001.(
mean	0.337662	0.328671	0.333666	55.920959	5.514486	15.471508	324.901678	2019-02-14 00:48:54.665334528	7.053946	309.4
min	0.000000	0.000000	0.000000	10.080000	1.000000	0.508500	10.678500	2019-01-01 00:00:00	4.000000	10.1
25%	0.000000	0.000000	0.000000	33.300000	3.000000	5.986500	125.716500	2019-01-24 00:00:00	5.500000	119.1
50%	0.000000	0.000000	0.000000	55.570000	5.000000	12.227500	256.777500	2019-02-13 00:00:00	6.900000	244.5
75%	1.000000	1.000000	1.000000	78.070000	8.000000	22.720500	477.130500	2019-03-08 00:00:00	8.500000	454.4
max	1.000000	1.000000	1.000000	99.960000	10.000000	49.650000	1042.650000	2019-03-30 00:00:00	97.000000	993.(
std	0.473149	0.469965	0.471758	26.385375	2.927121	11.716597	246.048534	NaN	3.324420	234.3

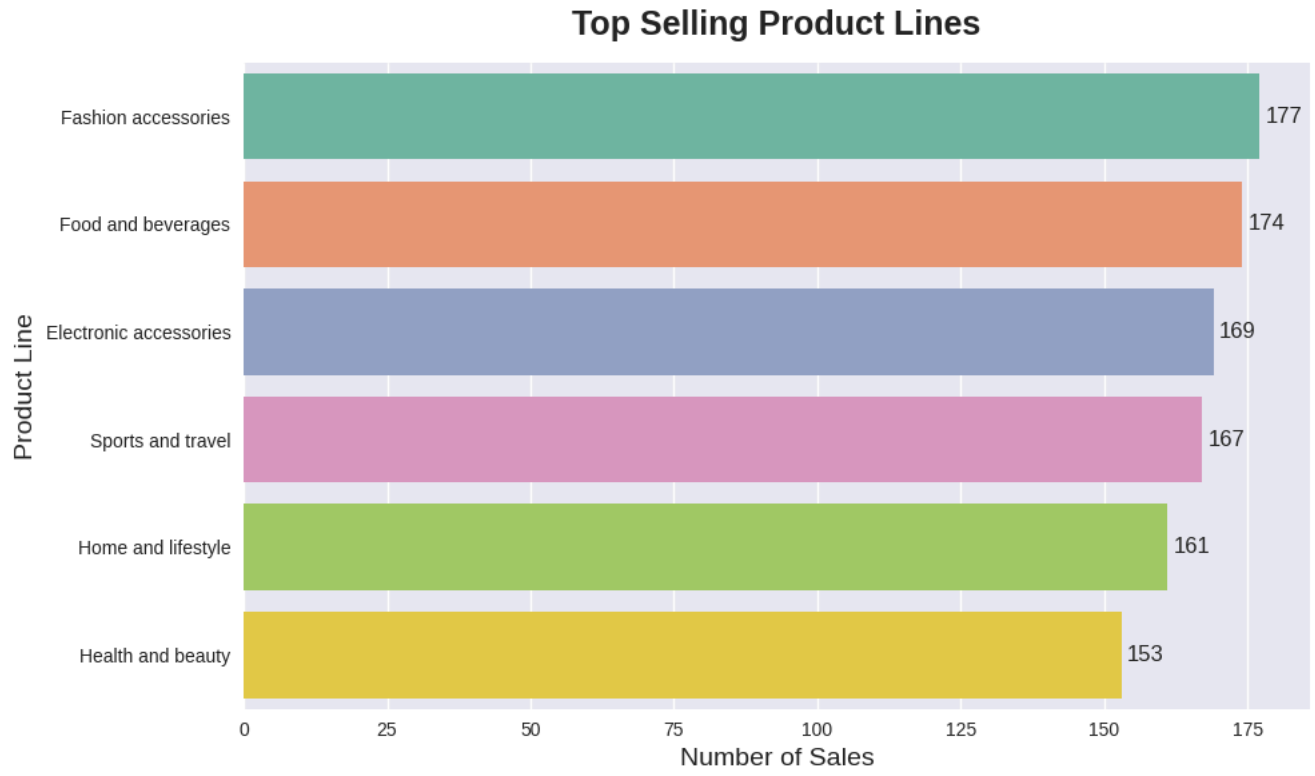
🏷️ 2. Product Line Analysis

Visualizing Which Product Lines Have The Highest Number Of Sales.

```
plt.figure(figsize=(10,6))

# Draw barplot with custom palette
ax = sns.countplot(
    y='Product line',
    data=df,
    order=df['Product line'].value_counts().index,
    palette="Set2"
)

# Add title and axis labels with bigger font
plt.title("Top Selling Product Lines", fontsize=18, fontweight='bold', pad=15)
plt.xlabel("Number of Sales", fontsize=14)
plt.ylabel("Product Line", fontsize=14)

# Add value labels next to each bar
for p in ax.patches:
    width = p.get_width()
```

```
    plt.text(width + 1,                # x-position
             p.get_y() + p.get_height()/2,  # y-position (middle of bar)
             int(width),               # the count value
             ha='left', va='center', fontsize=12)

# Remove right & top spines for cleaner look
sns.despine()

plt.tight_layout()
plt.show()
```

```
/tmp/ipython-input-2157281941.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `l

  ax = sns.countplot(
```



💡 **Insight:** Identify Which Product Lines Generate The Most Sales To Prioritize Stock.

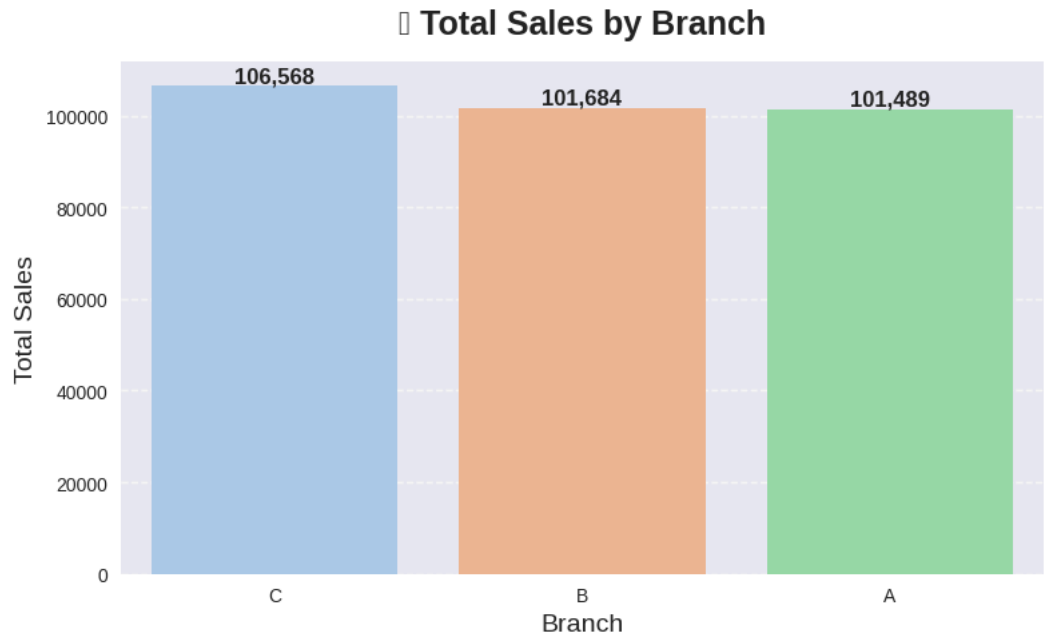
🏢 3. Branch Analysis

Compare Total Sales Across Different Branches.

```
plt.figure(figsize=(8,5))

# Sort branches by total sales
branch_totals = df.groupby('Branch')['Total_Calc'].sum().sort_values(ascending=False)

# Draw barplot with sorted order & custom colors
ax = sns.barplot(
    x=branch_totals.index,
    y=branch_totals.values,
    palette="pastel"
)

# Title and labels
plt.title("💰 Total Sales by Branch", fontsize=18, fontweight='bold', pad=15)
plt.xlabel("Branch", fontsize=14)
plt.ylabel("Total Sales", fontsize=14)

# Add value labels above each bar
for p in ax.patches:
```

```
    height = p.get_height()
    ax.annotate(f"{height:,.0f}",
                (p.get_x() + p.get_width() / 2., height),
                ha='center', va='bottom',
                fontsize=12, fontweight='bold')

# Remove top/right spines
sns.despine()

# Add light grid for better readability
plt.grid(axis='y', linestyle='--', alpha=0.6)

plt.tight_layout()
plt.show()
```

```
/tmp/ipython-input-1173283087.py:7: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `l

  ax = sns.barplot(
/tmp/ipython-input-1173283087.py:32: UserWarning: Glyph 128176 (\N{MONEY BAG}) missing from font(s) Liberation Sans.
  plt.tight_layout()
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128176 (\N{MONEY BAG}) missing from font
  fig.canvas.print_figure(bytes_io, **kw)
```



> 💡 **Insight:** Focus Marketing Or Promotions On Low-Performing Branches.

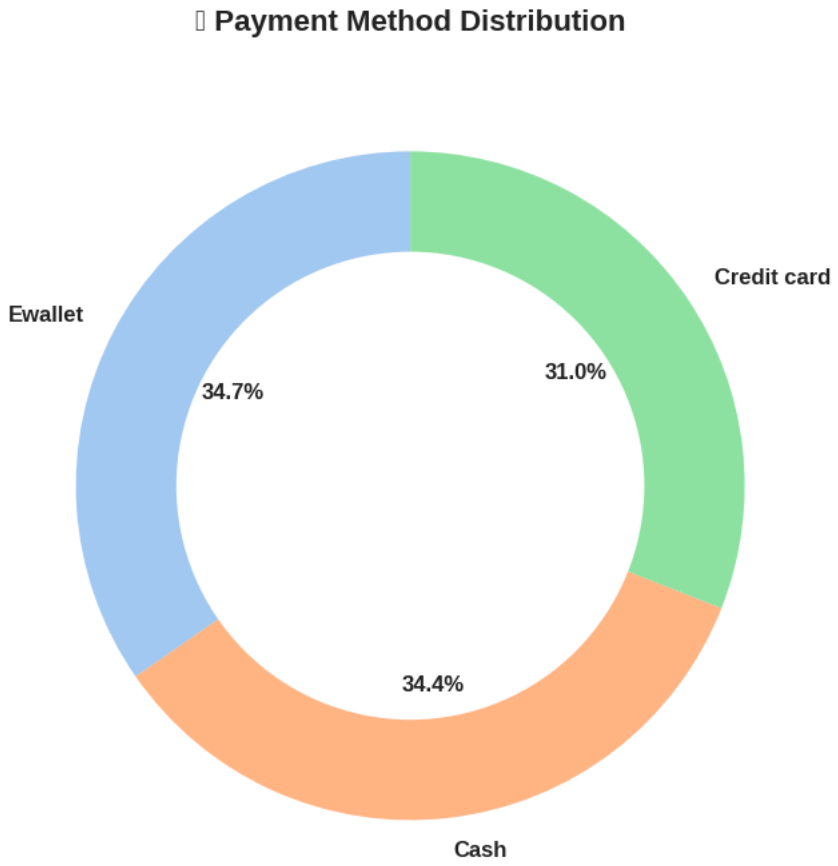
💳 4. Payment Method Analysis

Pie Chart Showing Preferred Payment Methods.

```
# 🎨Pie Chart for Payment Methods
payment_counts = df['Payment'].value_counts()

colors = sns.color_palette("pastel")  # Soft professional colors

plt.figure(figsize=(7,7))
plt.pie(payment_counts,
        labels=payment_counts.index,
        autopct='%1.1f%%',
        startangle=90,
        colors=colors,
        textprops={'fontsize': 12, 'weight': 'bold'})  # Make labels bold

# Add a central circle to make it a "donut chart"
centre_circle = plt.Circle((0,0),0.70,fc='white')
fig = plt.gcf()
```

```
fig.gca().add_artist(centre_circle)

plt.title("💳 Payment Method Distribution", fontsize=16, fontweight='bold', pad=20)
plt.tight_layout()
plt.show()
```

```
/tmp/ipython-input-231497083.py:20: UserWarning: Glyph 128179 (\N{CREDIT CARD}) missing from font(s) Liberation Sans.
  plt.tight_layout()
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128179 (\N{CREDIT CARD}) missing from fo
  fig.canvas.print_figure(bytes_io, **kw)
```



> 💡 **Insight:** Consider Promoting Alternative Payment Methods Or Offering Discounts

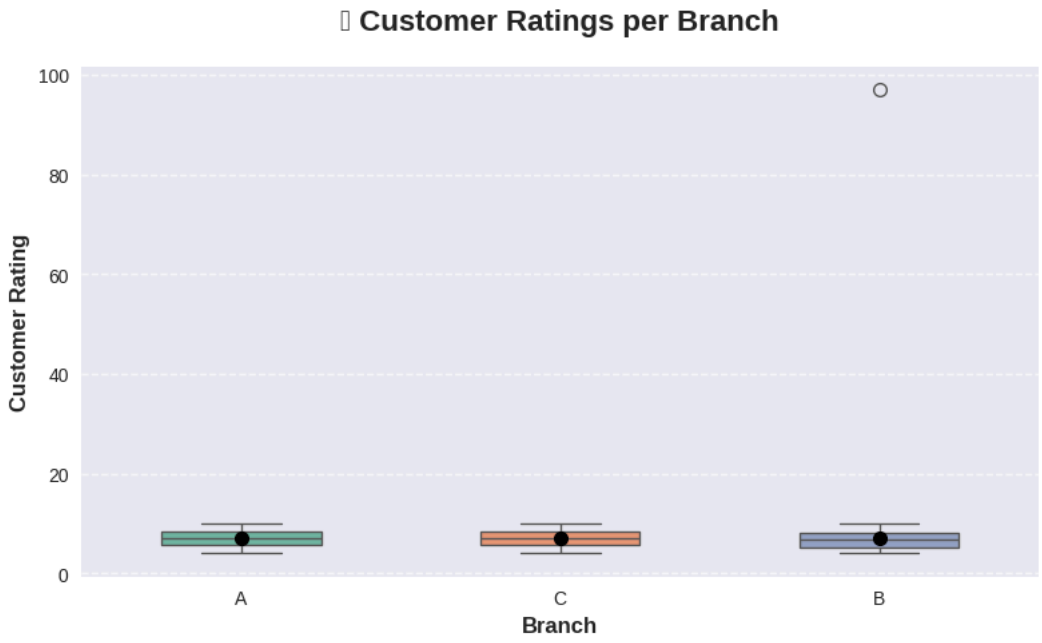
⭐ 5. Customer Rating Analysis

Distribution Of Customer Ratings Per Branch.

```
#Boxplot for Customer Ratings per Branch
plt.figure(figsize=(8,5))

# Modern & clean style boxplot
sns.boxplot(
    x='Branch',
    y='Rating',
    data=df,
    palette='Set2',        # Professional pastel colors
    width=0.5,             # Slimmer boxplot for a cleaner look
    showmeans=True,        # Show mean as a special marker
    meanprops={"marker":"o",
               "markerfacecolor":"black",
               "markeredgecolor":"black",
               "markersize":"8"}  # Style the mean point
)

# Add title and labels
plt.title("⭐ Customer Ratings per Branch", fontsize=16, fontweight='bold', pad=20)
```

```
plt.xlabel("Branch", fontsize=12, fontweight='bold')
plt.ylabel("Customer Rating", fontsize=12, fontweight='bold')

# Add grid for better readability
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.tight_layout()
plt.show()
```

```
/tmp/ipython-input-1916959012.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `l

  sns.boxplot(
/tmp/ipython-input-1916959012.py:26: UserWarning: Glyph 11088 (\N{WHITE MEDIUM STAR}) missing from font(s) Liberation Sans.
  plt.tight_layout()
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 11088 (\N{WHITE MEDIUM STAR}) missing fr
  fig.canvas.print_figure(bytes_io, **kw)
```



> 💡 **Insight:** Low-Rating Branches Might Need Service Quality Improvements.

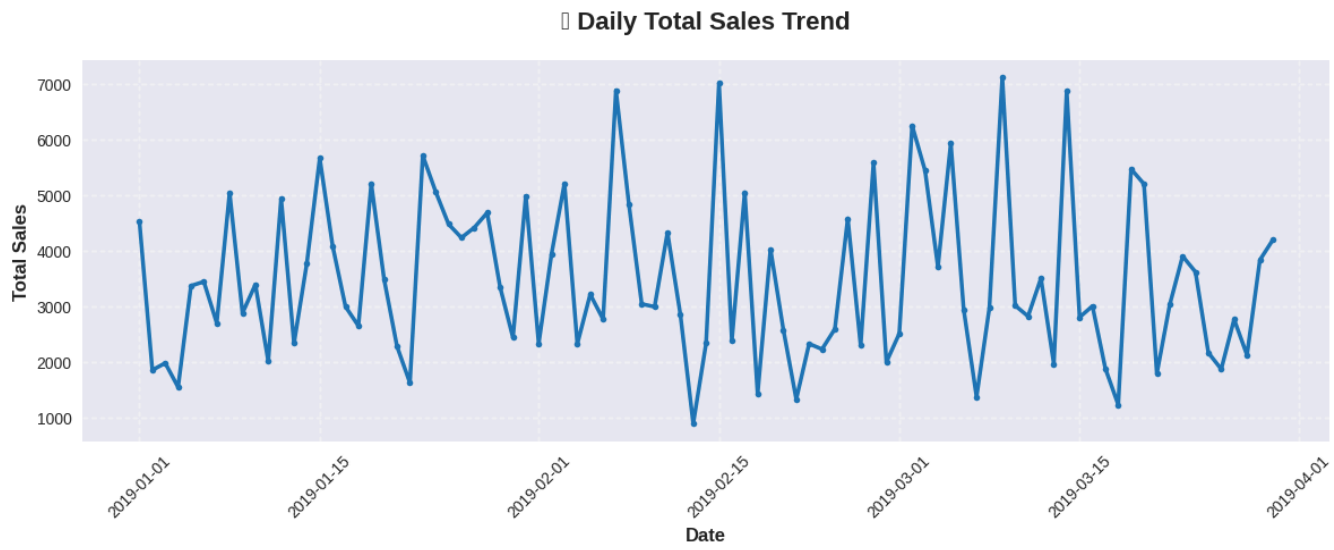
⏳ 6. Time-Based Analysis

Analyze Daily And Hourly Sales To Find Peak Periods.

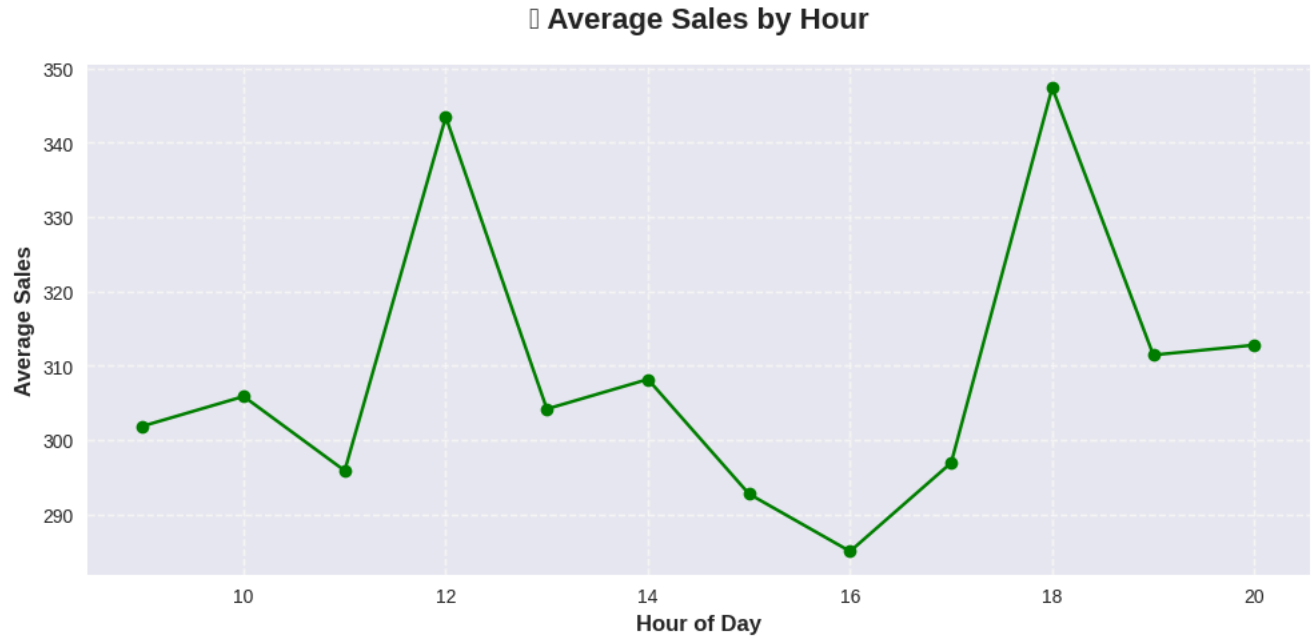
```
# 📈 DAILY SALES TREND
sales_by_date = df.groupby('Date')['Total_Calc'].sum()

plt.figure(figsize=(12,5))
plt.plot(sales_by_date.index, sales_by_date.values, color='tab:blue', linewidth=2.5, marker='o', markersize=4)

plt.title("📈 Daily Total Sales Trend", fontsize=16, fontweight='bold', pad=20)
plt.xlabel("Date", fontsize=12, fontweight='bold')
plt.ylabel("Total Sales", fontsize=12, fontweight='bold')
plt.grid(alpha=0.3, linestyle="--")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
# 📊 ================================
# Average Sales by Hour
# ================================
plt.figure(figsize=(10,5))
df.groupby('Hour')['Total_Calc'].mean().plot(kind='line', marker='o', color='green')
plt.title("💰 Average Sales by Hour", fontsize=16, fontweight='bold', pad=20)
plt.xlabel("Hour of Day", fontsize=12, fontweight='bold')
plt.ylabel("Average Sales", fontsize=12, fontweight='bold')
plt.grid(linestyle='--', alpha=0.6)
```

```
plt.tight_layout()
plt.show()
```

```
/tmp/ipython-input-3800578542.py:12: UserWarning: Glyph 128200 (\N{CHART WITH UPWARDS TREND}) missing from font(s) Liberation Sans.
  plt.tight_layout()
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128200 (\N{CHART WITH UPWARDS TREND}) mi
  fig.canvas.print_figure(bytes_io, **kw)
```



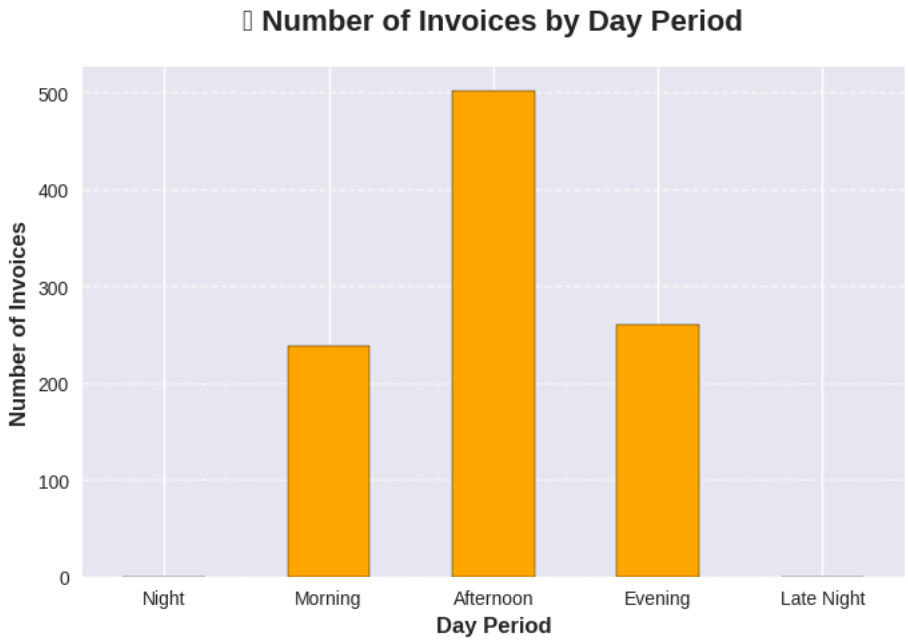

```
/tmp/ipython-input-3800578542.py:23: UserWarning: Glyph 128176 (\N{MONEY BAG}) missing from font(s) Liberation Sans.
  plt.tight_layout()
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128176 (\N{MONEY BAG}) missing from font
  fig.canvas.print_figure(bytes_io, **kw)
```



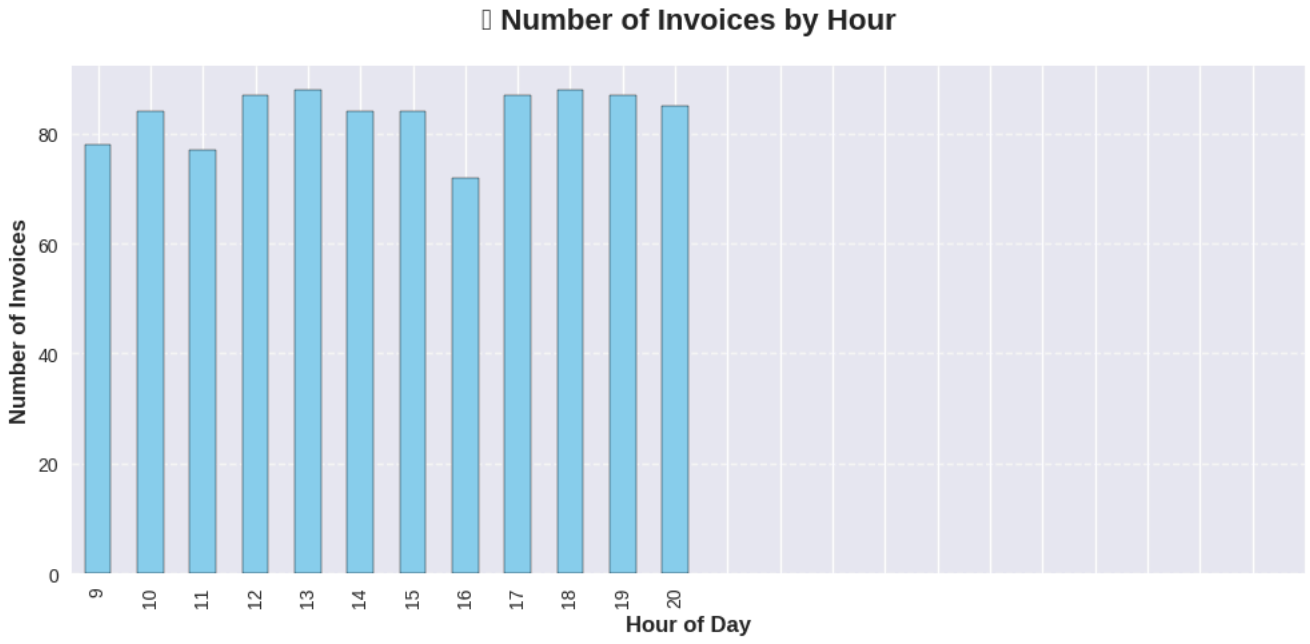

```
# 📊 ================================
# Sales by Day Period
# ================================
plt.figure(figsize=(7,5))
# تأكد إن الترتيب يكون منطقي
day_order = ['Night', 'Morning', 'Afternoon', 'Evening', 'Late Night']
df['Day_Period'].value_counts().reindex(day_order).dropna().plot(kind='bar', color='orange', edgecolor='black')
plt.title("🌞 Number of Invoices by Day Period", fontsize=16, fontweight='bold', pad=20)
plt.xlabel("Day Period", fontsize=12, fontweight='bold')
plt.ylabel("Number of Invoices", fontsize=12, fontweight='bold')
plt.xticks(rotation=0)
```

```
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
# 📊 ================================
# Sales by Hour (Count of Invoices)
# ================================
plt.figure(figsize=(10,5))
df['Hour'].value_counts().sort_index().plot(kind='bar', color='skyblue', edgecolor='black')
plt.title("⏰ Number of Invoices by Hour", fontsize=16, fontweight='bold', pad=20)
plt.xlabel("Hour of Day", fontsize=12, fontweight='bold')
plt.ylabel("Number of Invoices", fontsize=12, fontweight='bold')
plt.xticks(range(0, 24))
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

```
/tmp/ipython-input-3314397932.py:13: UserWarning: Glyph 127774 (\N{SUN WITH FACE}) missing from font(s) Liberation Sans.
  plt.tight_layout()
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 127774 (\N{SUN WITH FACE}) missing fro
  fig.canvas.print_figure(bytes_io, **kw)
```




```
/tmp/ipython-input-3314397932.py:25: UserWarning: Glyph 9200 (\N{ALARM CLOCK}) missing from font(s) Liberation Sans.
  plt.tight_layout()
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 9200 (\N{ALARM CLOCK}) missing from fo
  fig.canvas.print_figure(bytes_io, **kw)
```



> ℹ️ Note: Missing Hour values were imputed before analysis to ensure complete insights

💡 **Insight:** Use daily sales patterns to forecast demand, optimize inventory levels, and schedule targeted marketing campaigns on high-traffic days.

📊 Key Metrics Calculation

Below we calculate the most important KPIs that will be used in the Business Insights section.

```
# 📊 Key Metrics Calculation

# 1️⃣ Total Sales by Product Line
print("🔝 Total Sales by Product Line:")
print(df.groupby('Product line')['Total_Calc'].sum().sort_values(ascending=False))
print("-" * 50)

# 2️⃣ Total Sales by Branch
print("🏙️ Total Sales by Branch:")
print(df.groupby('Branch')['Total_Calc'].sum().sort_values(ascending=False))
print("-" * 50)

# 3️⃣ Payment Method Distribution
print("💳 Payment Method Distribution:")
print(df['Payment'].value_counts())
print("-" * 50)

# 4️⃣ Total Sales by Hour
print("⏰ Total Sales by Hour:")
print(df.groupby('Hour')['Total_Calc'].sum().sort_values(ascending=False))
```

```
🔝 Total Sales by Product Line:
Product line
Food and beverages        53696.39
Sports and travel         52530.91
Electronic accessories    52410.55
Fashion accessories       52136.25
Home and lifestyle        51819.69
Health and beauty         47145.81
Name: Total_Calc, dtype: float64
--------------------------------------------------
🏙️ Total Sales by Branch:
Branch
C    106567.54
B    101683.50
A    101488.56
Name: Total_Calc, dtype: float64
--------------------------------------------------
💳 Payment Method Distribution:
Payment
Ewallet        347
Cash           344
Credit card    310
Name: count, dtype: int64
--------------------------------------------------
⏰ Total Sales by Hour:
Hour
18    30571.25
12    29881.39
19    27094.04
13    26766.22
20    26584.98
14    25885.08
17    25831.41
10    25691.39
15    24588.78
9     23544.15
11    22780.85
16    20520.06
Name: Total_Calc, dtype: float64
```

🔍 Business Insights & Recommendations

Key Findings:

- **Top Product Lines: Food & Beverages** generate the highest sales (53,696.39), followed closely by **Sports & Travel** and **Electronic Accessories**.
- **Best Branch: Branch C** leads with the highest total sales (106,567.54), slightly outperforming **B** and **A**.
- **Payment Preference: E-wallet** is the most popular payment method (347 transactions), indicating a digital-first customer base.
- **Peak Hours: 16:00 (4 PM)** and **18:00 (6 PM)** are the strongest sales hours, showing evening shopping spikes.

Recommendations:

1. **Stock Prioritization:** Allocate more inventory to **Food & Beverages** and **Sports & Travel** to avoid stockouts.